HOSTEL ATTENDANCE SYSTEM

URL:
https://docs.google.com/spreadsheets/d/1ittSBj3nteDDFAQgeOKvLw_D7ZV3HQFXF_hU0CC7juU/edit?gid=941624154#gid=941624154

```
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <HTTPClient.h>
#include <Adafruit_Fingerprint.h>
// ========================================================
// WIFI DETAILS
// ========================================================
const char* WIFI_SSID = "Starlit";
const char* WIFI_PASSWORD = "password";
const char* GOOGLE_SCRIPT_URL =
"https://script.google.com/macros/s/AKfycby8cjjWZnPgG0j8RCbSX8BNpOXSQh2XGdjl2ChphhZA3ihKvvRG-Il0XVJJjpEaSQn4/exec";
// ========================================================
// R307S AND BUZZER CONNECTIONS
// ========================================================
// R307S TX -> ESP32 GPIO 16
// R307S RX -> ESP32 GPIO 17
// R307S VCC -> suitable power supply
// R307S GND -> ESP32 GND
//
// Active buzzer signal -> ESP32 GPIO 25
// Buzzer GND -> ESP32 GND

#define BUZZER_PIN 25

HardwareSerial fingerSerial(2);
Adafruit_Fingerprint finger = Adafruit_Fingerprint(&fingerSerial);
```

```
// ========================================================
// STUDENT DETAILS
// ========================================================

struct Student
{
  const char* name;
  const char* roomNo;
  const char* blockNo;
  const char* mobileNo;
};

// ========================================================
// STUDENT DATABASE: FINGERPRINT IDs 5 TO 13
// ========================================================

Student students[14] =
{
  {"", "", "", ""},
  {"Student 1", "101", "A", "0000000000"},
  {"Student 2", "102", "A", "0000000000"},
  {"Student 3", "201", "B", "0000000000"},
  {"Student 4", "202", "B", "0000000000"},
  {"Gopika", "31", "C", "6369464303"},
  {"Raji", "22", "C", "8610775116"},
  {"Sandhiya", "35", "C", "9360012592"},
  {"Thirisha", "22", "D", "8778284820"},
  {"amina", "35", "C", "6381750572"},
  {"vaishu", "33", "C", "6385634558"},
  {"sandhiya.a", "27", "C", "7010099211"},
  {"rakshantha", "27", "C", "7904903003"},
  {"Indhuja", "31", "C", "7305522735"}
```

```
};

// ================================================
// BUZZER: SUCCESS ONLY
// Active buzzer: two short beeps
// ================================================

void buzzerSuccess()
{
  digitalWrite(BUZZER_PIN, HIGH);
  delay(150);
  digitalWrite(BUZZER_PIN, LOW);

  delay(100);

  digitalWrite(BUZZER_PIN, HIGH);
  delay(150);
  digitalWrite(BUZZER_PIN, LOW);
}

// ================================================
// SETUP
// ================================================

void setup()
{
  Serial.begin(115200);

  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  fingerSerial.begin(57600, SERIAL_8N1, 16, 17);
```

```
delay(1000);

Serial.println();
Serial.println("==========================================");
Serial.println("       HOSTEL FINGERPRINT SYSTEM");
Serial.println("==========================================");

// ================================================
// CONNECT TO WIFI
// ================================================

Serial.println("Connecting to Wi-Fi...");

WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

int wifiAttempt = 0;

while (WiFi.status() != WL_CONNECTED && wifiAttempt < 30)
{
  delay(500);
  Serial.print(".");
  wifiAttempt++;
}

Serial.println();

if (WiFi.status() == WL_CONNECTED)
{
  Serial.println("Wi-Fi connected!");
  Serial.print("ESP32 IP Address: ");
  Serial.println(WiFi.localIP());
```

```
}
else
{
  Serial.println("Wi-Fi connection failed!");
  Serial.println("Fingerprint system will continue.");
}

// ================================================
// FINGERPRINT SENSOR
// ================================================

finger.begin(57600);

if (finger.verifyPassword())
{
  Serial.println("R307S fingerprint sensor found!");
}
else
{
  Serial.println("R307S NOT FOUND!");
  Serial.println("Check VCC, GND, TX and RX.");

  while (true)
  {
    delay(100);
  }
}

finger.getTemplateCount();

Serial.print("Fingerprints stored in sensor: ");
Serial.println(finger.templateCount);
```

```
  showMenu();
}

// ========================================================
// MAIN LOOP
// ========================================================

void loop()
{
  if (Serial.available())
  {
    char choice = Serial.read();

    while (Serial.available())
    {
      Serial.read();
    }

    if (choice == '1')
    {
      enrollMenu();
    }
    else if (choice == '2')
    {
      identifyFingerprint();
    }
    else if (choice == '3')
    {
      showStudents();
    }
    else
```

```
    {
      Serial.println("Invalid choice.");
    }

    delay(500);
    showMenu();
  }
}

// ================================================
// MAIN MENU
// ================================================

void showMenu()
{
  Serial.println();
  Serial.println("========================================");
  Serial.println("              MAIN MENU");
  Serial.println("========================================");

  Serial.println("1. Register / Upload Fingerprint");
  Serial.println("2. Identify Student");
  Serial.println("3. Show Registered Students");

  Serial.println("========================================");
  Serial.println("Enter your choice:");
}

// ================================================
// SHOW STUDENTS
// ================================================
```

```
void showStudents()
{
  Serial.println();
  Serial.println("=========================================");
  Serial.println("        REGISTERED STUDENTS");
  Serial.println("=========================================");

  for (int i = 5; i <= 13; i++)
  {
    Serial.println();

    Serial.print("Fingerprint ID : ");
    Serial.println(i);

    Serial.print("Name           : ");
    Serial.println(students[i].name);

    Serial.print("Room Number    : ");
    Serial.println(students[i].roomNo);

    Serial.print("Block Number   : ");
    Serial.println(students[i].blockNo);

    Serial.print("Mobile Number  : ");
    Serial.println(students[i].mobileNo);

    Serial.println("----------------------------------------");
  }
}

// ========================================================
// ENROLL MENU
```

```
// ================================================

void enrollMenu()
{
  Serial.println();
  Serial.println("========================================");
  Serial.println("          REGISTER STUDENT");
  Serial.println("========================================");

  for (int i = 5; i <= 13; i++)
  {
    Serial.print(i);
    Serial.print(" -> ");
    Serial.println(students[i].name);
  }

  Serial.println("Enter Fingerprint ID (5 to 13):");

  while (!Serial.available())
  {
    delay(100);
  }

  int id = Serial.parseInt();

  while (Serial.available())
  {
    Serial.read();
  }

  if (id < 5 || id > 13)
  {
```

```
    Serial.println("Invalid ID. Enter ID from 5 to 13.");
    return;
  }

  Serial.println();
  Serial.print("Name          : ");
  Serial.println(students[id].name);

  Serial.print("Room Number   : ");
  Serial.println(students[id].roomNo);

  Serial.print("Block Number  : ");
  Serial.println(students[id].blockNo);

  Serial.print("Mobile Number : ");
  Serial.println(students[id].mobileNo);

  enrollFingerprint(id);
}

// ========================================================
// ENROLL FINGERPRINT
// No buzzer during registration
// ========================================================

bool enrollFingerprint(uint8_t id)
{
  int p = -1;

  Serial.println("Place finger on the sensor...");

  while (p != FINGERPRINT_OK)
```

```
{
  p = finger.getImage();

  if (p == FINGERPRINT_NOFINGER)
  {
    delay(100);
  }
  else if (p == FINGERPRINT_OK)
  {
    Serial.println("Fingerprint image captured.");
  }
  else
  {
    Serial.print("Fingerprint error: ");
    Serial.println(p);
    return false;
  }
}

p = finger.image2Tz(1);

if (p != FINGERPRINT_OK)
{
  Serial.println("Could not convert first fingerprint.");
  return false;
}

Serial.println("First fingerprint converted.");
Serial.println("Remove your finger.");

delay(2000);
```

```
while (finger.getImage() != FINGERPRINT_NOFINGER)
{
  delay(100);
}

Serial.println("Place the SAME finger again...");

p = -1;

while (p != FINGERPRINT_OK)
{
  p = finger.getImage();

  if (p == FINGERPRINT_NOFINGER)
  {
    delay(100);
  }
  else if (p == FINGERPRINT_OK)
  {
    Serial.println("Second fingerprint captured.");
  }
  else
  {
    Serial.print("Fingerprint error: ");
    Serial.println(p);
    return false;
  }
}

p = finger.image2Tz(2);

if (p != FINGERPRINT_OK)
```

```
{
  Serial.println("Could not convert second fingerprint.");
  return false;
}

Serial.println("Creating fingerprint model...");

p = finger.createModel();

if (p != FINGERPRINT_OK)
{
  Serial.println("Fingerprints did NOT match.");
  return false;
}

Serial.println("Fingerprints matched.");
Serial.println("Saving fingerprint to sensor memory...");

p = finger.storeModel(id);

if (p == FINGERPRINT_OK)
{
  Serial.println("REGISTRATION SUCCESSFUL!");

  Serial.print("Fingerprint ID : ");
  Serial.println(id);

  Serial.print("Student Name   : ");
  Serial.println(students[id].name);

  // No buzzer during registration
  return true;
```

```
  }
  else
  {
    Serial.println("Failed to store fingerprint.");
    return false;
  }
}

// ========================================================
// SEND ATTENDANCE TO GOOGLE SHEETS
// ========================================================

void sendAttendanceToGoogle(
  int id,
  const char* name,
  const char* room,
  const char* block,
  const char* mobile,
  const char* status)
{
  if (WiFi.status() != WL_CONNECTED)
  {
    Serial.println("Wi-Fi not connected.");
    Serial.println("Attendance cannot be sent to Google Sheet.");
    return;
  }

  Serial.println("Sending attendance to Google Sheet...");

  WiFiClientSecure client;
  client.setInsecure();
```

```
HTTPClient http;

String url = String(GOOGLE_SCRIPT_URL);

url += "?id=";
url += String(id);

url += "&name=";
url += String(name);

url += "&room=";
url += String(room);

url += "&block=";
url += String(block);

url += "&mobile=";
url += String(mobile);

url += "&status=";
url += String(status);

Serial.println("Connecting to Google...");

if (http.begin(client, url))
{
  http.setFollowRedirects(HTTPC_STRICT_FOLLOW_REDIRECTS);

  int httpCode = http.GET();

  Serial.print("HTTP Response Code: ");
  Serial.println(httpCode);
```

```
    if (httpCode > 0)
    {
      String response = http.getString();

      Serial.print("Google Response: ");
      Serial.println(response);

      if (response.indexOf("Attendance saved successfully") >= 0)
      {
        Serial.println("GOOGLE SHEET UPDATE SUCCESSFUL");
      }
      else
      {
        Serial.println("Google confirmation was not received.");
      }
    }
    else
    {
      Serial.println("Failed to send attendance.");
    }

    http.end();
  }
  else
  {
    Serial.println("Unable to connect to Google Script.");
  }
}
// ====================================================
// IDENTIFY FINGERPRINT
// Buzzer only for a registered fingerprint match
```

```
// ================================================
void identifyFingerprint()
{
  Serial.println();
  Serial.println("==========================================");
  Serial.println("        STUDENT IDENTIFICATION");
  Serial.println("==========================================");
  Serial.println("Place your finger on the sensor...");

  int p = -1;
  // GET FINGERPRINT IMAGE
  while (p != FINGERPRINT_OK)
  {
    p = finger.getImage();
    if (p == FINGERPRINT_NOFINGER)
    {
      delay(100);
    }
    else if (p == FINGERPRINT_OK)
    {
      Serial.println("Fingerprint detected.");
    }
    else
    {
      Serial.println("Error reading fingerprint.");
      return; // No beep
    }
  }
  // CONVERT IMAGE
  p = finger.image2Tz();
  if (p != FINGERPRINT_OK)
  {
```

```
  Serial.println("Could not convert fingerprint.");
  return; // No beep
}
// SEARCH FOR MATCH
p = finger.fingerFastSearch();
// FINGERPRINT FOUND
if (p == FINGERPRINT_OK)
{
  int id = finger.fingerID;
  Serial.print("Fingerprint ID : ");
  Serial.println(id);
  if (id >= 5 && id <= 13)
  {
    // BEEP ONLY FOR A REGISTERED MATCH
    buzerSuccess();
    Serial.println("STUDENT FOUND");
    Serial.print("Student Name   : ");
    Serial.println(students[id].name);
    Serial.print("Room Number    : ");
    Serial.println(students[id].roomNo);
    Serial.print("Block Number   : ");
    Serial.println(students[id].blockNo);
    Serial.print("Mobile Number  : ");
    Serial.println(students[id].mobileNo);
    Serial.print("Confidence     : ");
    Serial.println(finger.confidence);
    // SEND ATTENDANCE TO GOOGLE SHEETS
    sendAttendanceToGoogle(
      id,
      students[id].name,
      students[id].roomNo,
      students[id].blockNo,
```

```
      students[id].mobileNo,
      "VERIFIED"
    );
    // SERIAL ATTENDANCE DATA
    Serial.print("ATTENDANCE,");
    Serial.print(id);
    Serial.print(",");
    Serial.print(students[id].name);
    Serial.print(",");
    Serial.print(students[id].roomNo);
    Serial.print(",");
    Serial.print(students[id].blockNo);
    Serial.print(",");
    Serial.print(students[id].mobileNo);
    Serial.println(",VERIFIED");
  }
  else
  {
    Serial.println("Student details not registered.");
    // No beep
  }
}
// FINGERPRINT NOT FOUND
else if (p == FINGERPRINT_NOTFOUND)
{
  Serial.println("FINGERPRINT NOT FOUND");
  Serial.println("Student is not registered.");
  // No beep
}
// OTHER SEARCH ERROR
else
{
```

```
    Serial.print("Fingerprint search error. Code: ");
    Serial.println(p);
    // No beep
  }

  // WAIT FOR FINGER REMOVAL

  Serial.println("Remove your finger.");

  while (finger.getImage() != FINGERPRINT_NOFINGER)
  {
    delay(100);
  }

  delay(1000);
}
```